

RUTHVIK VISHWANATHWAR

Senior Data Engineer • ruthvikvishwa@gmail.com • 919-397-3265 • ruthvikvishwanathwar.com

SUMMARY

Senior Data Engineer and **Databricks Certified Data Engineer** with 11 years of experience designing enterprise-scale lakehouse architectures that enable analytics, compliance, and AI-driven decision-making across healthcare and financial services.

CERTIFICATIONS

Databricks: Databricks Certified Data Engineer Associate, Platform Architect, Certified Platform Administrator, Generative AI Fundamentals, AI Agent Fundamentals, Lakehouse Fundamentals | dbt fundamentals | SQL for Data Science

PROFESSIONAL EXPERIENCE

Senior Data Engineer | *Frisco, TX (Client/s; Sentara Health, Georgia Pacific)*

May 2022 – Present

- Strategized and implemented migration of 80+ legacy SSIS/SQL Server ETL workflows to a Medallion Delta Lake architecture using PySpark, Spark SQL, and Azure Databricks; reduced pipeline runtime by 60% and cut production incidents tied to legacy job failures.
- Designed and implemented a reconciliation framework covering field mapping, crosswalk logic, and configurable tolerance thresholds in PySpark and Delta Lake, decreasing false-positive discrepancies by over 60% and eliminating manual review escalations to engineering.
- Architected a lineage and exception-tracking layer within Delta Lake to convert the reconciliation database into a reusable ML-ready asset, enabling the data science team to build drift detection on top of production data quality history.
- Led performance and cost tuning across enterprise-scale Delta tables using Z-ordering, partitioning, OPTIMIZE/MERGE, and cluster policy design; reduced query times by up to 70% and brought a 14-hour batch job under 3 hours.
- Established governance and CI/CD standards for HIPAA/PHI pipelines using Unity Catalog, RBAC, tokenization, and masking; mentored a team of 15 developers on Databricks workflows, new ETL pipeline patterns post-migration, and AI-assisted development practices to accelerate delivery.

Stack: Azure Databricks, PySpark, Spark SQL, ADF, Delta Lake, Delta Live Tables, SQL Server 2019, Azure Synapse, Power BI, SSIS, Python

Senior Data Engineer | *Frisco, TX (Client/s; Blue Cross Blue Shield of Kansas)*

May 2021 – May 2022

- Designed and implemented validation and reconciliation checks across eligibility, claims, and provider-metric pipelines in SQL Server and Databricks, identifying missing records, duplicate claims, and inconsistent member attributes prior to executive reporting.
- Designed orchestration architecture across Azure Data Factory, Databricks Jobs, and Airflow DAGs, standardizing retry logic, dependency handling, SLA monitoring, and backfill operations across high-volume claims data.
- Optimized Databricks workload efficiency through partition strategy, file compaction, and cluster configuration standards, reducing compute cost and improving pipeline scalability across enterprise datasets.
- Refactored high-volume SQL workloads through indexing strategy, execution-plan analysis, and stored procedure redesign, reducing report refresh time by 40% for finance and claims operational reporting.

Stack: SQL Server 2016/2017, Power BI, Power Query, DAX, GitHub Actions, SSMS, JIRA, Confluence, Databricks, Airflow DAGs

Database Developer | *Addison, TX (Client/s; Bank of America)*

Nov 2019 – Apr 2021

- Designed the SSIS extraction and transformation architecture for multi-source ingestion across Excel, flat files, and relational databases, choosing Conditional Split, Fuzzy Lookup, and Multicast over simpler load patterns once source data quality made a straight load unreliable.
- Standardized the ETL approach across multiple operational banking systems that had grown independently, cutting down on the one-off scripts each system previously relied on for consistency checks.
- Traced recurring reporting slowdowns to specific stored procedures and multi-join queries, and resolved them by rewriting execution plans rather than adding indexes reactively, which became the debugging approach the team adopted afterward.

Stack: SQL Server 2016/2017, SSIS, SSRS, SSAS, Power BI, Visual Studio 2017, JIRA, Confluence, Bitbucket

Database Developer | *Highlands Ranch, CO (Client/s; Visa Inc, Evolent Health, Vintech)*

Aug 2016 – Oct 2019

- Maintained referential integrity and data consistency using SQL constraints; reviewed and refactored complex multi-join queries to improve performance across high-transaction financial datasets.
- Designed automated ETL processes ensuring end-to-end data consistency, quality, and scalability across multiple operational systems.

Stack: SQL Server 2016, SSIS, SSRS, Visual Studio 2017, JIRA, Confluence, Bitbucket

SKILLS

- **Cloud, Platforms & Engineering:** Azure Data Factory, Databricks, Synapse Analytics, ADLS Gen2, Blob Storage, Logic Apps, DevOps, Microsoft Fabric, OneLake, Snowflake, Power BI, ETL/ELT Pipelines, Delta Lake, Medallion Architecture, Data Lakehouse, Data Warehousing, SSIS Modernization, Spark Structured Streaming, Lakehouse Federation, Airflow, DAG Design, SLA Monitoring, Backfills, Dependency Management, dbt.
- **Programming & Optimization:** Python, PySpark, SQL, T-SQL, DAX, Spark SQL, Z-Ordering, Compaction, Caching, OPTIMIZE/MERGE, Data Skipping, Index Tuning, Query Tuning, Cost Optimization.
- **AI/ML, DevOps & Governance:** Generative AI, LLM Data Pipelines, MLflow, Feature Engineering, AI Agent Orchestration, Git, GitHub, Azure DevOps CI/CD, GitHub Actions, Databricks Notebooks, JIRA, Confluence, ServiceNow, HIPAA/PHI Masking, Access Controls, Schema Enforcement, Audit Logging, SCDs, Data Quality Validation.

EDUCATION

M.S. Information Technology | GPA: 3.8/4.00 | Campbellsville University, Kentucky

M.S. Computer Science | GPA: 3.7/4.00 | New York Institute of Technology, Manhattan, NY

B.S. Information Technology | GPA: 3.5/4.00 | Scient Institute of Technology, India